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KYOTO UNIVERSITY

Statistical Learning Theory - Classification -

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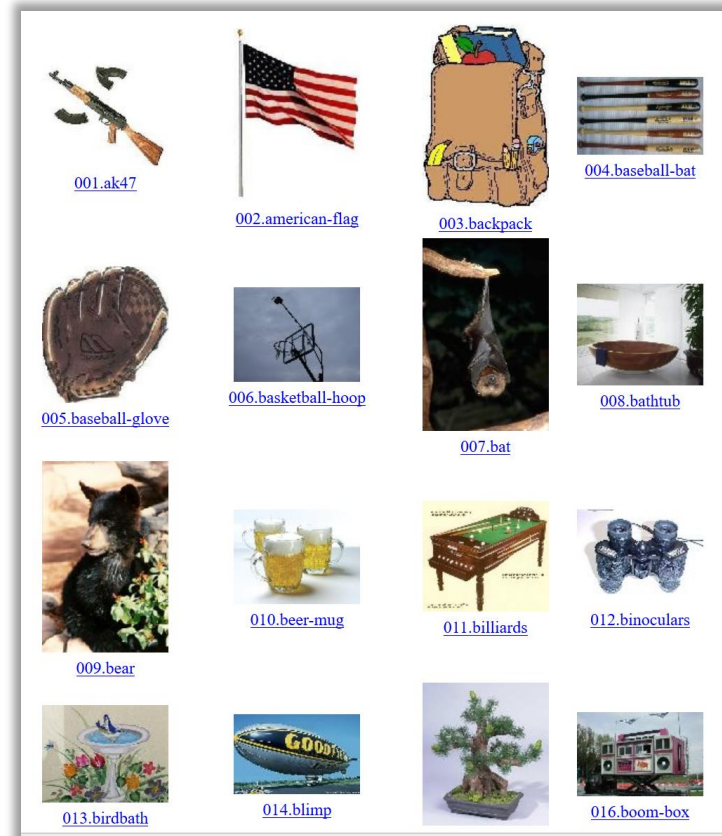
DEPARTMENT OF INTELLIGENCE SCIENCE
AND TECHNOLOGY

Classification

Classification:

Supervised learning for predicting discrete variable

- Goal: Obtain a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ ($\mathcal{Y}$: discrete domain)
 - E.g. $x \in \mathcal{X}$ is an image and $y \in \mathcal{Y}$ is the type of object appearing in the image
 - Two-class classification: $\mathcal{Y} = \{+1, -1\}$
- Training dataset:
 N pairs of an input and an output
 $\{(\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(N)}, y^{(N)})\}$



http://www.vision.caltech.edu/Image_Datasets/Caltech256/

Some applications of classification:

From binary to multi-class classification

- Binary (two-class) classification:
 - Purchase prediction: Predict if a customer $\mathbf{x}$ will buy a particular product (+1) or not (-1)
 - Credit risk prediction: Predict if an obligor $\mathbf{x}$ will pay back a debt (+1) or not (-1)
- Multi-class classification ($\neq$ Multi-label classification):
 - Text classification: Categorize a document $\mathbf{x}$ into one of several categories, e.g., {politics, economy, sports, ...}
 - Image classification: Categorize the object in an image $\mathbf{x}$ into one of several object names, e.g., {AK5, American flag, backpack, ...}
 - Action recognition: Recognize the action type ({running, walking, sitting, ...}) that a person is taking from sensor data $\mathbf{x}$

A simple model for classification:

Linear classifier

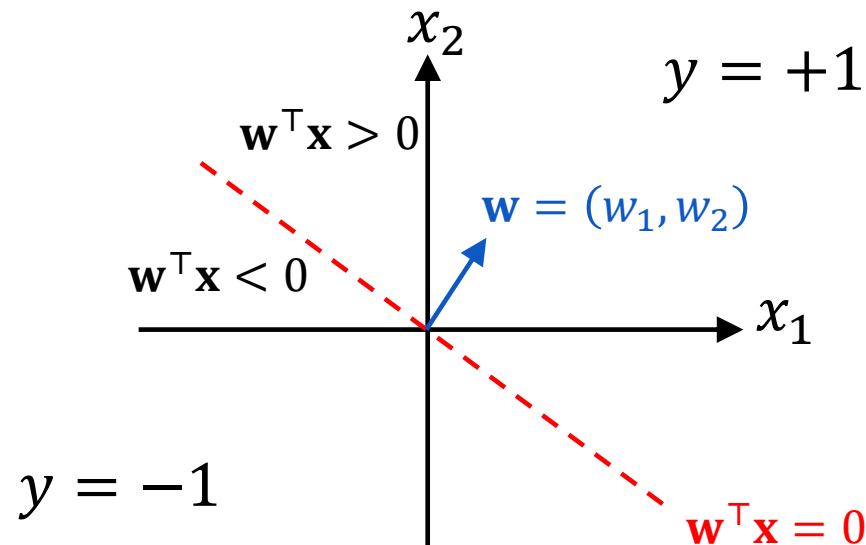
- Linear (binary) classification model:

$$y = \text{sign}(\mathbf{w}^\top \mathbf{x}) = \text{sign}(w_1 x_1 + w_2 x_2 + \cdots + w_D x_D)$$

– $|\mathbf{w}^\top \mathbf{x}|$ indicates the intensity of belief

– $\mathbf{w}^\top \mathbf{x} = 0$ gives a separating hyperplane

- $\mathbf{w}$: normal vector perpendicular to the separating hyperplane



Learning framework:

Loss minimization and statistical estimation

- Two learning frameworks

1. Loss minimization: $L(\mathbf{w}) = \sum_{i=1}^N \ell(y^{(i)}, \mathbf{w}^\top \mathbf{x}^{(i)})$

- Loss function ℓ : directly handles utility of predictions
- Regularization term $R(\mathbf{w})$

2. Statistical estimation (likelihood maximization):

$$L(\mathbf{w}) = \prod_{i=1}^N f_{\mathbf{w}}(y^{(i)} | \mathbf{x}^{(i)})$$

- Probabilistic model: generation process of class labels
- Prior distribution $P(\mathbf{w})$

- They are often equivalent : $\begin{cases} \text{Loss} = \text{Probabilistic model} \\ \text{Regularization} = \text{Prior} \end{cases}$

Classification problem in loss minimization framework:

Minimize loss function + regularization term

- Minimization problem: $\mathbf{w}^* = \operatorname{argmin}_{\mathbf{w}} L(\mathbf{w}) + R(\mathbf{w})$
 - Loss function $L(\mathbf{w})$: Fitness to training data
 - Regularization term $R(\mathbf{w})$: Penalty on the model complexity to avoid overfitting to training data (usually norm of $\mathbf{w}$)
- Loss function should reflect the number of misclassifications on training data
 - Zero-one loss seems reasonable:

$$\ell^{(i)}(y^{(i)}, \mathbf{w}^\top \mathbf{x}^{(i)}) = \begin{cases} 0 & (y^{(i)} = \operatorname{sign}(\mathbf{w}^\top \mathbf{x}^{(i)})) \\ 1 & (y^{(i)} \neq \operatorname{sign}(\mathbf{w}^\top \mathbf{x}^{(i)})) \end{cases}$$

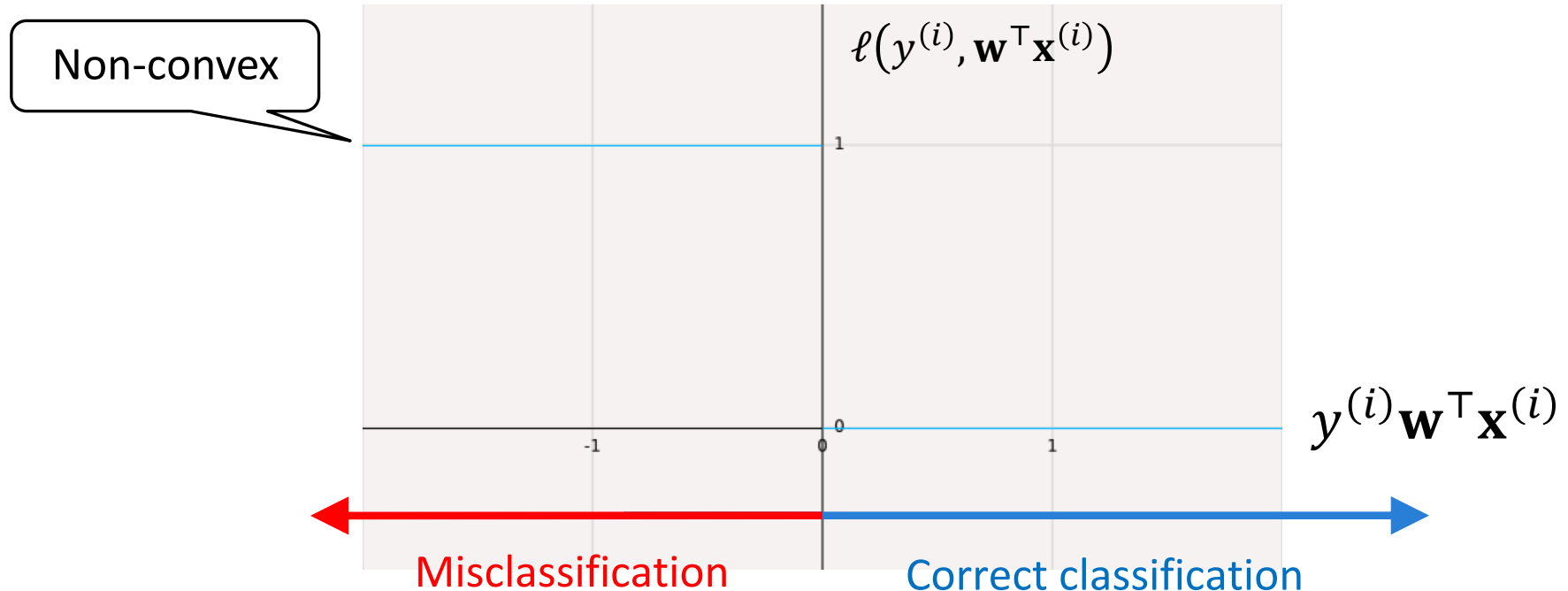
Correct classification

Incorrect classification

Zero-one loss:

Number of misclassification is hard to minimize

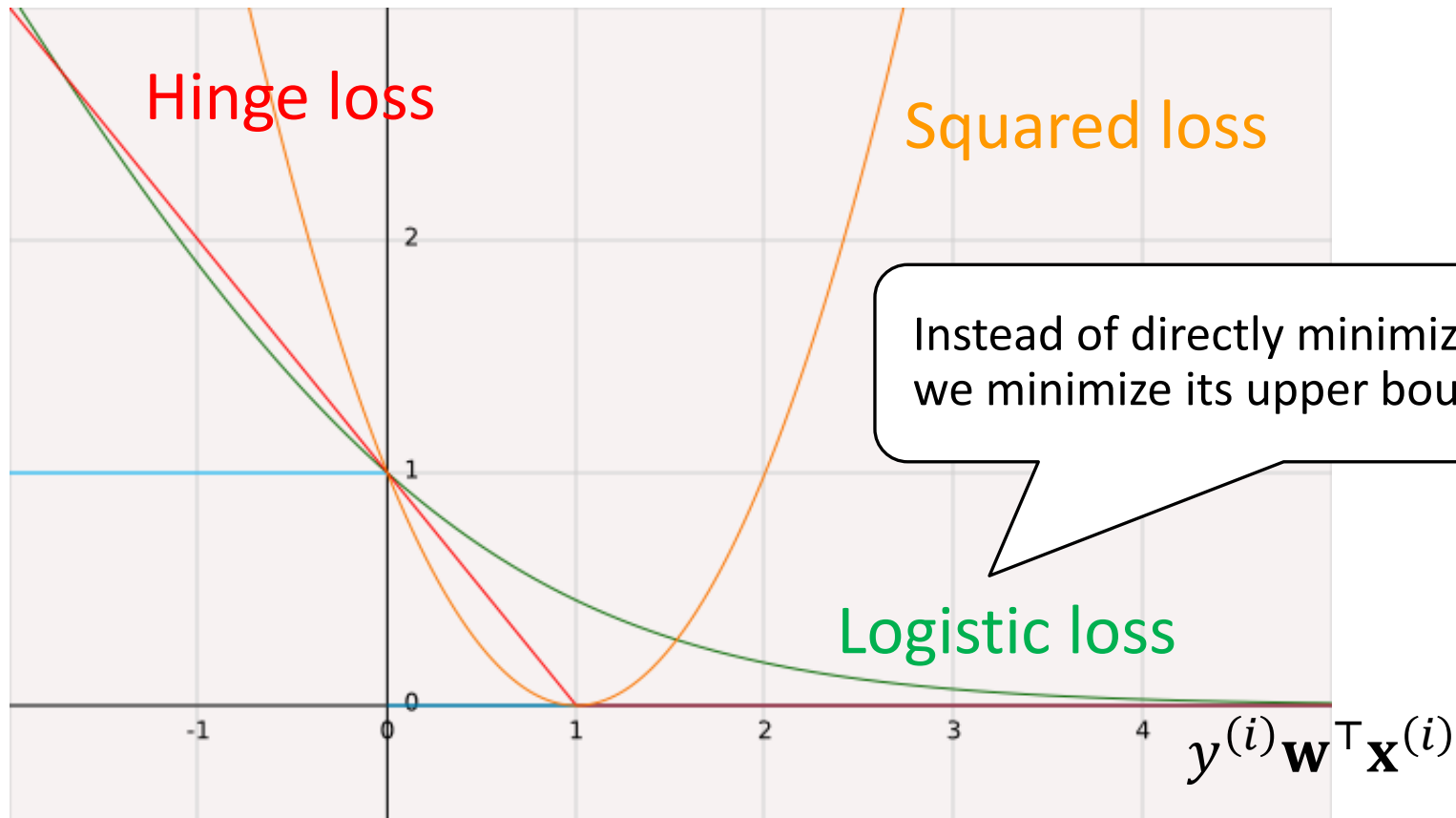
- Zero-one loss: $\ell(y^{(i)}, \mathbf{w}^\top \mathbf{x}^{(i)}) = \begin{cases} 0 & (y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} > 0) \\ 1 & (y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)} \leq 0) \end{cases}$
- Non-convex function is hard to optimize directly



Convex surrogates of zero-one loss:

Different functions lead to different learning machines

- Convex surrogates: Upper bounds of zero-one loss
 - Hinge loss → SVM, Logistic loss → logistic regression, ...



Logistic regression

Logistic regression:

Minimization of logistic loss is a convex optimization

- Logistic loss:

$$\ell(y^{(i)}, \mathbf{w}^\top \mathbf{x}^{(i)}) = \frac{1}{\ln 2} \ln(1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)}))$$

- (Regularized) Logistic regression:

$$\mathbf{w}^* = \operatorname{argmin}_{\mathbf{w}} \sum_{i=1}^N \ln(1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)})) + \lambda \|\mathbf{w}\|_2^2$$

Convex 😊



Statistical interpretation:

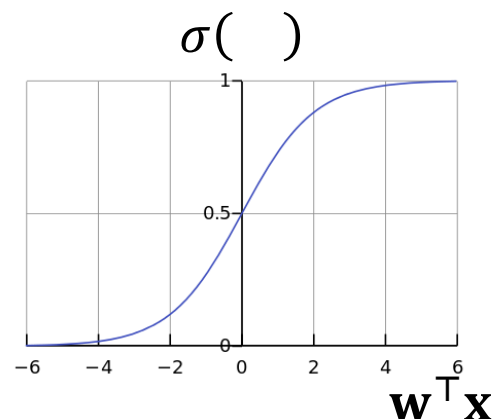
Logistic loss min. as MLE of logistic regression model

- *Minimization of logistic loss is equivalent to maximum likelihood estimation of logistic regression model*

- Logistic regression model (conditional probability):

$$f_{\mathbf{w}}(y = +1|\mathbf{x}) = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{x})}$$

- σ : Logistic function ($\sigma: \mathbb{R} \rightarrow (0,1)$)



- Log likelihood:

$$L(\mathbf{w}) = \sum_{i=1}^N \log f_{\mathbf{w}}(y^{(i)}|\mathbf{x}^{(i)}) = - \sum_{i=1}^N \log(1 + \exp(-y^{(i)} \mathbf{w}^T \mathbf{x}))$$

a.k.a. Cross-entropy

$$\left(= \sum_{i=1}^N \delta(y^{(i)} = 1) \log \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{x})} + \delta(y^{(i)} = -1) \log \left(1 - \frac{1}{1 + \exp(-\mathbf{w}^T \mathbf{x})} \right) \right)$$

Parameter estimation of logistic regression :

Numerical nonlinear optimization

- Objective function of (regularized) logistic regression:

$$L(\mathbf{w}) = \sum_{i=1}^N \ln(1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)})) + \lambda \|\mathbf{w}\|_2^2$$

- Minimization of logistic loss / MLE of logistic regression model has no closed form solution
- Numerical nonlinear optimization methods are used
 - Iterate parameter updates: $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} + \mathbf{d}$ (until convergence)



Parameter update :

Find the best update minimizing the objective function

- By update $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} + \mathbf{d}$, the objective function will be:

$$L_{\mathbf{w}}(\mathbf{d}) = \sum_{i=1}^N \ln(1 + \exp(-y^{(i)}(\mathbf{w} + \mathbf{d})^\top \mathbf{x}^{(i)})) + \lambda \|\mathbf{w} + \mathbf{d}\|_2^2$$

- Find $\mathbf{d}^*$ that minimizes $L_{\mathbf{w}}(\mathbf{d})$:

- $\mathbf{d}^* = \operatorname{argmin}_{\mathbf{d}} L_{\mathbf{w}}(\mathbf{d})$

- ... but so far, this problem has not been made easier at all ... 🤔

Finding the best parameter update :

Approximate the objective with Taylor expansion

- Taylor expansion:

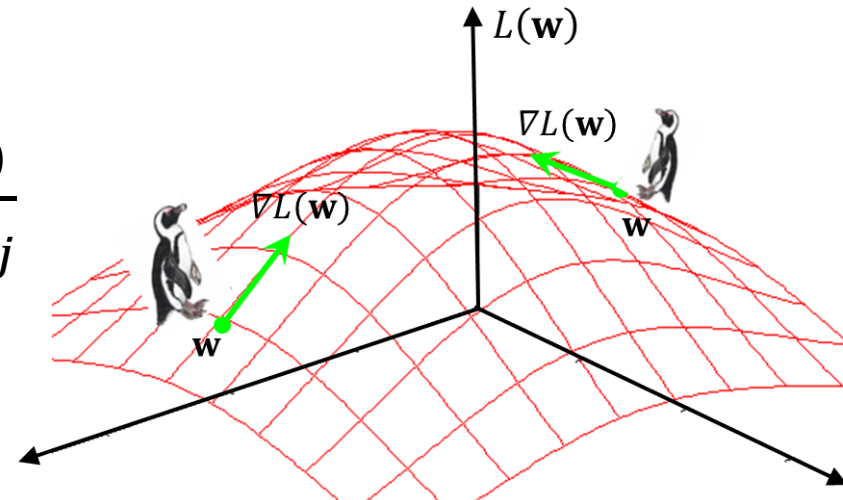
$$L_{\mathbf{w}}(\mathbf{d}) = L(\mathbf{w}) + \mathbf{d}^\top \nabla L(\mathbf{w}) + \frac{1}{2} \mathbf{d}^\top \mathbf{H}(\mathbf{w}) \mathbf{d} + O(\mathbf{d}^3)$$

3rd-order term

- Gradient vector: $\nabla L(\mathbf{w}) = \left(\frac{\partial L(\mathbf{w})}{\partial w_1}, \frac{\partial L(\mathbf{w})}{\partial w_2}, \dots, \frac{\partial L(\mathbf{w})}{\partial w_D} \right)^\top$

- Steepest direction

- Hessian matrix: $[H(\mathbf{w})]_{i,j} = \frac{\partial^2 L(\mathbf{w})}{\partial w_i \partial w_j}$



Newton update :

Minimizes the second order approximation

- Approximated Taylor expansion (neglecting the 3rd order term):

$$L_{\mathbf{w}}(\mathbf{d}) \approx L(\mathbf{w}) + \mathbf{d}^\top \nabla L(\mathbf{w}) + \frac{1}{2} \mathbf{d}^\top \mathbf{H}(\mathbf{w}) \mathbf{d} + \cancel{\mathcal{O}(\mathbf{d}^3)}$$

- Derivative w.r.t. $\mathbf{d}$: $\frac{\partial L_{\mathbf{w}}(\mathbf{d})}{\partial \mathbf{d}} \approx \nabla L(\mathbf{w}) + \mathbf{H}(\mathbf{w}) \mathbf{d}$
- Setting it to be $\mathbf{0}$, we obtain $\mathbf{d} = -\mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$
- Newton update formula:

$$\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$$



Modified Newton update:

Second order approximation + linear search

- The correctness of the update $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$ depends on the second-order approximation:

$$L_{\mathbf{w}}(\mathbf{d}) \approx L(\mathbf{w}) + \mathbf{d}^{\top} \nabla L(\mathbf{w}) + \frac{1}{2} \mathbf{d}^{\top} \mathbf{H}(\mathbf{w}) \mathbf{d}$$

—This is not actually true for most cases

- Use only the direction of $\mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$ and update with $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \eta \mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$
- Learning rate $\eta > 0$ is determined by linear search:
$$\eta^* = \operatorname{argmin}_{\eta} L(\mathbf{w} - \eta \mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w}))$$

(Steepest) gradient descent:

Simple update without computing inverse Hessian

- Computing **the inverse of Hessian matrix** is costly

- Newton update: $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \eta \mathbf{H}(\mathbf{w})^{-1} \nabla L(\mathbf{w})$

- (Steepest) gradient descent:

- Replacing $\mathbf{H}(\mathbf{w})^{-1}$ with $\mathbf{I}$ gives

$$\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \eta \nabla L(\mathbf{w})$$

Gradient of
objective function

- $\nabla L(\mathbf{w})$ is the steepest direction
- Learning rate η is determined by line search



Summary:

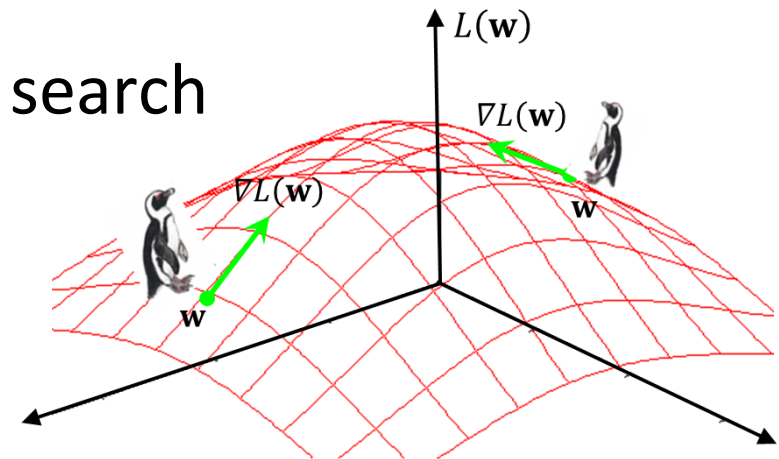
Gradient descent

- Steepest gradient descent is the simplest optimization method:
- Update the parameter in the steepest direction of the objective function

$$\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \eta \nabla L(\mathbf{w})$$

– Gradient: $\nabla L(\mathbf{w}) = \left(\frac{\partial L(\mathbf{w})}{\partial w_1}, \frac{\partial L(\mathbf{w})}{\partial w_2}, \dots, \frac{\partial L(\mathbf{w})}{\partial w_D} \right)^T$

– Learning rate η is determined by line search or just fixed at a small value



Example of gradient descent: Gradient of logistic regression

- $L(\mathbf{w}) = \sum_{i=1}^N \ln(1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)}))$

- $$\begin{aligned} \frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} &= \sum_{i=1}^N \frac{1}{1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)})} \frac{\partial (1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)}))}{\partial \mathbf{w}} \\ &= - \sum_{i=1}^N \frac{1}{1 + \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)})} \exp(-y^{(i)} \mathbf{w}^\top \mathbf{x}^{(i)}) y^{(i)} \mathbf{x}^{(i)} \\ &= - \sum_{i=1}^N (1 - f_{\mathbf{w}}(y^{(i)} | \mathbf{x}^{(i)})) y^{(i)} \mathbf{x}^{(i)} \end{aligned}$$

Can be easily computed with the current prediction probabilities

Mini batch optimization:

Efficient training using data subsets

- Objective function for N instances:

$$L(\mathbf{w}) = \sum_{i=1}^N \ell(\mathbf{w}^\top \mathbf{x}^{(i)}) + \lambda R(\mathbf{w})$$

- Its derivative $\frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} = \sum_{i=1}^N \frac{\partial \ell(\mathbf{w}^\top \mathbf{x}^{(i)})}{\partial \mathbf{w}} + \lambda \frac{\partial R(\mathbf{w})}{\partial \mathbf{w}}$ needs $O(N)$ computation

- Approximate this with only one instance:

$$\frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} \approx N \frac{\partial \ell(\mathbf{w}^\top \mathbf{x}^{(j)})}{\partial \mathbf{w}} + \lambda \frac{\partial R(\mathbf{w})}{\partial \mathbf{w}} \quad (\text{Stochastic approximation})$$

- Also we can do this with $1 < M < N$ instances:

$$\frac{\partial L(\mathbf{w})}{\partial \mathbf{w}} \approx \frac{N}{M} \sum_{j \in \text{MiniBatch}} \frac{\partial \ell(\mathbf{w}^\top \mathbf{x}^{(j)})}{\partial \mathbf{w}} + \lambda \frac{\partial R(\mathbf{w})}{\partial \mathbf{w}} \quad (\text{Mini batch})$$

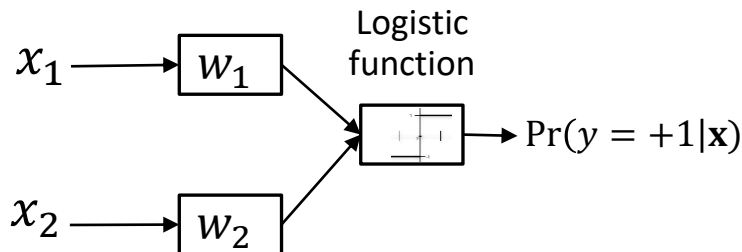
From Logistic Regression to Neural Networks

Neural networks:

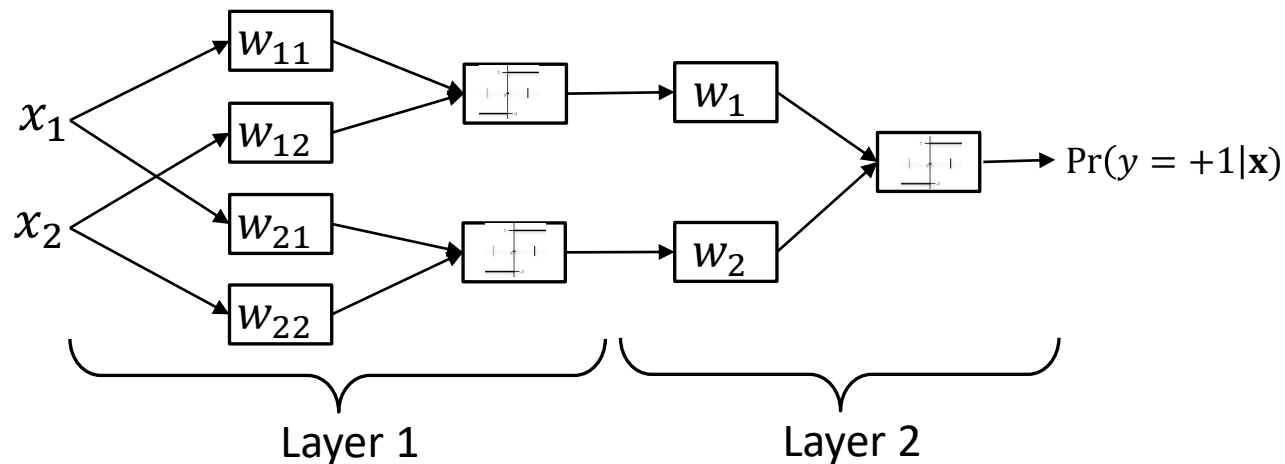
Multi-layered logistic regression models

- (Very roughly speaking,) a neural network can be seen as multi-layered logistic regression models
 - Outputs of some logistic regression models are inputs to other logistic regression models
- The form of final output is still $\Pr(y = +1|\mathbf{x})$

Logistic regression



Neural network (2 layers)

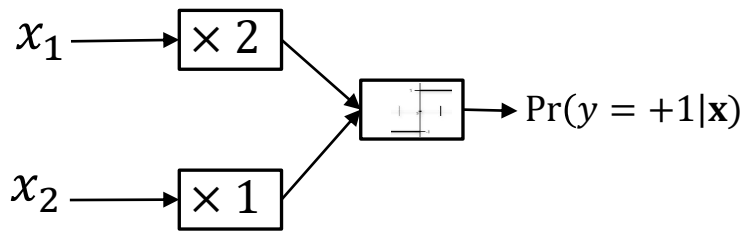


Why do we need to stack layers?

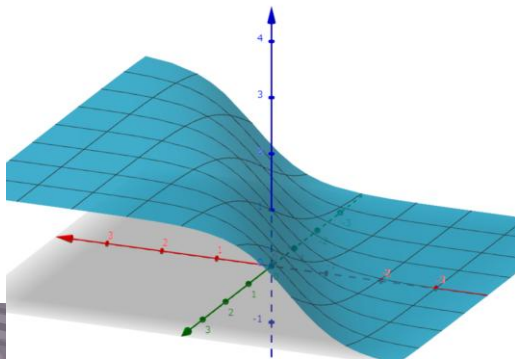
Nonlinear classification

- (1-layer) logistic regression only allow linear classification (AND/OR)
- Gains *non-linear* expressive power by stacking two layers (XOR)
- Universal approximation theorem: neural network can represent arbitrary functions by introducing many intermediate units

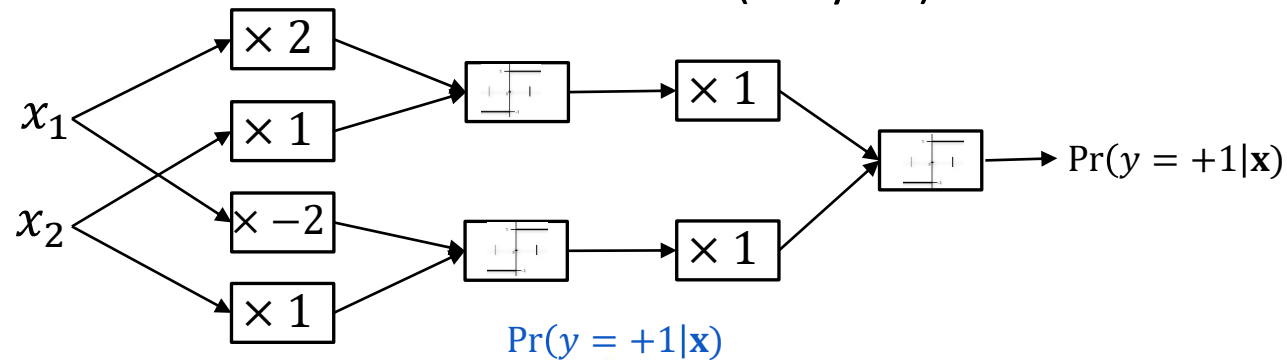
Logistic regression



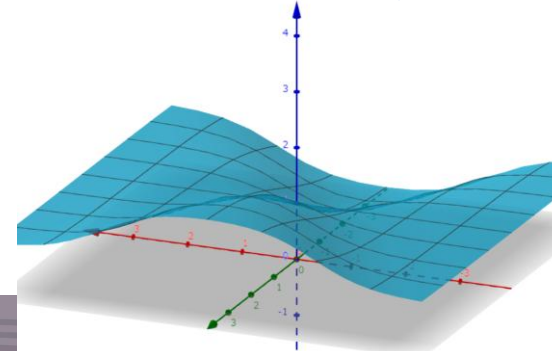
$\Pr(y = +1|\mathbf{x})$



Neural network (2 layers)



$\Pr(y = +1|\mathbf{x})$



Parameter estimation for neural networks:

Gradient is all you need

- The objective function for training a neural network is the same as that of logistic regression, i.e., the *log likelihood*:

– Stochastic approximation for the log likelihood for the i -th instance:

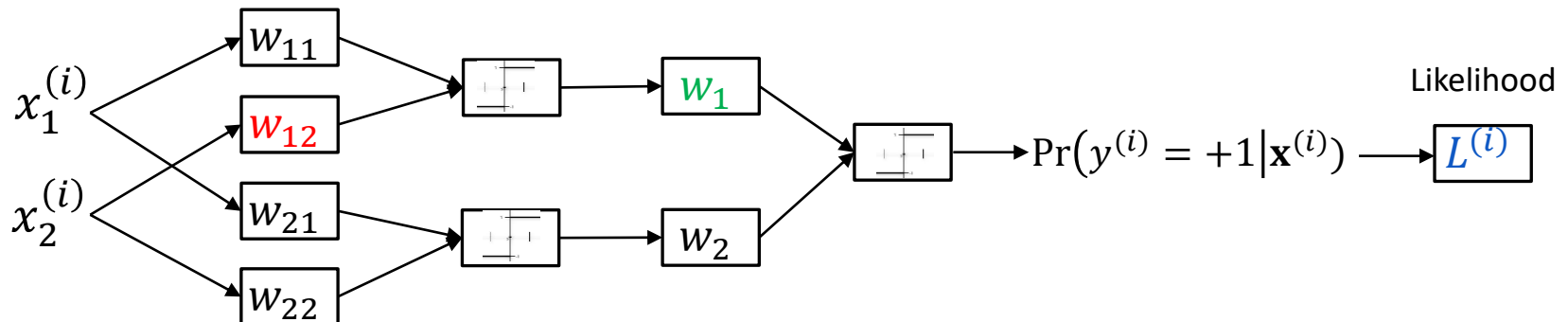
$$L^{(i)}(\mathbf{w}) = \delta(y^{(i)} = 1) \log f_{\mathbf{w}}(y^{(i)} = +1 | \mathbf{x}^{(i)}) + \delta(y^{(i)} = -1) \log f_{\mathbf{w}}(y^{(i)} = -1 | \mathbf{x}^{(i)})$$

– Update : $\mathbf{w}^{\text{NEW}} \leftarrow \mathbf{w} - \eta \nabla L^{(i)}(\mathbf{w})$

a.k.a. Cross-entropy

- How can we obtain the gradient $\nabla L^{(i)}(\mathbf{w}) = \partial L^{(i)} / \partial \mathbf{w}$?

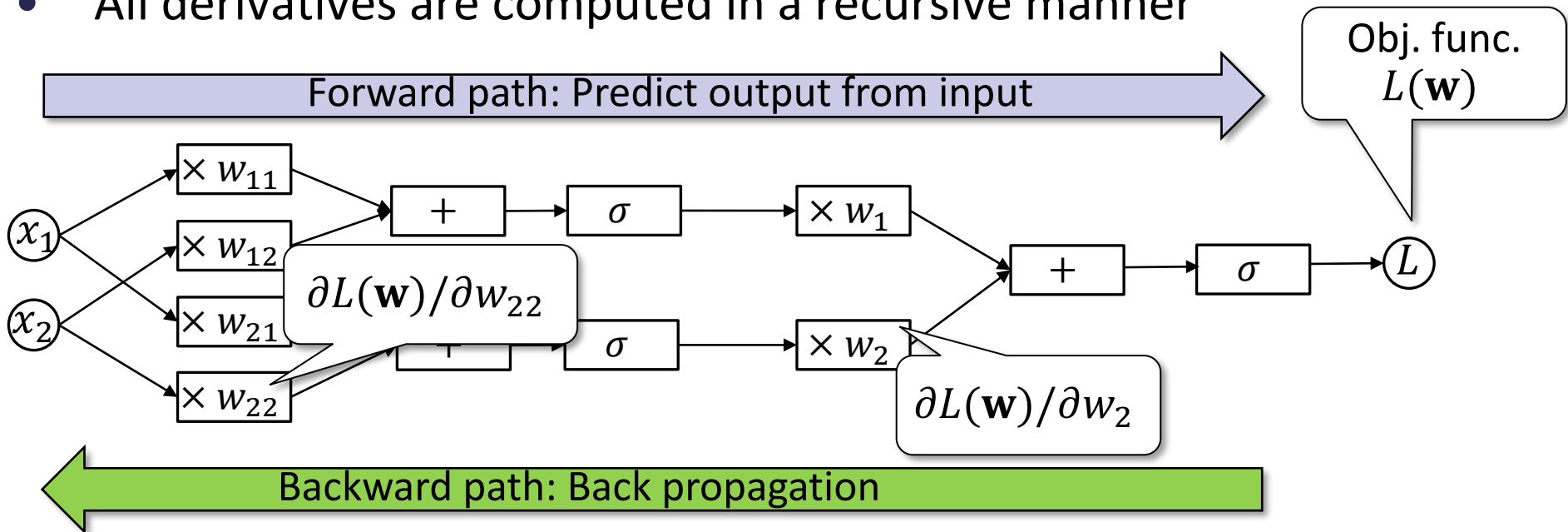
– E.g., how can we obtain $\partial L^{(i)} / \partial w_{12}$? (They are separated via w_1)



Back propagation:

An efficient strategy to compute the gradient

- Once we differentiate the objective function, we can apply SA
- *Back-propagation* can efficiently compute all the gradients
 - Forward path: computes output (objective func.) from the input
 - Backward path: computes derivatives from output to input
- All derivatives are computed in a recursive manner



Multi-class Classification

Multi-class classification:

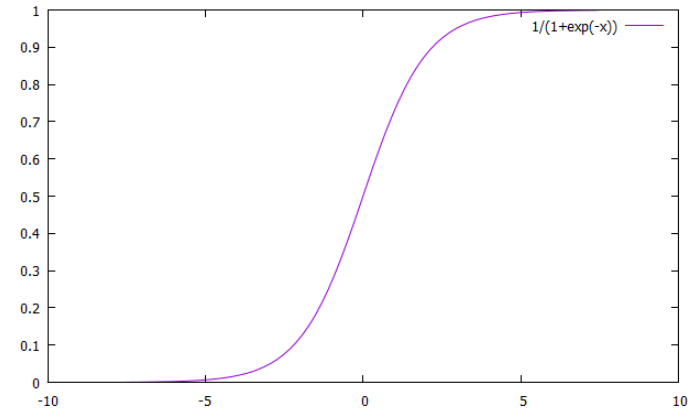
Generalization of supervised two-class classification

- Training dataset: $\{ (\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(i)}, y^{(i)}), \dots, (\mathbf{x}^{(N)}, y^{(N)}) \}$
 - input $\mathbf{x}^{(i)} \in \mathcal{X} = \mathbb{R}^D$: D -dimensional real vector
 - output $y^{(i)} \in \mathcal{Y}$: one-dimensional scalar
- Estimate a *deterministic mapping* $f: \mathcal{X} \rightarrow \mathcal{Y}$ (often with a confidence value) or a *conditional probability* $P(y|\mathbf{x})$
- Classification
 - $\mathcal{Y} = \{+1, -1\}$: Two-class classification
 - $\mathcal{Y} = \{1, 2, \dots, K\}$: K -class multi-class classification
 - hand-written digit recognition, text classification, ...

Two-class classification model:

One model with one parameter vector

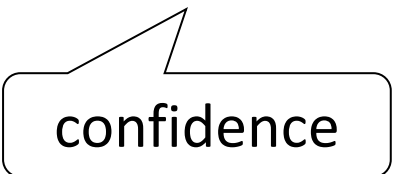
- Two-class classification model
 - Linear classifier: $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x}) \in \{+1, -1\}$
 - Logistic regression: $P(y|\mathbf{x}) = \frac{1}{1 + \exp(-\mathbf{w}^\top \mathbf{x})}$
 - The model is specified by a parameter vector $\mathbf{w} = (w_1, w_2, \dots, w_D)^\top$



Simple approaches to multi-class classification:

Reduction to two-class classification

- Reduction to a set of two-class classification problems
- Approach 1: One-versus-rest
 - Construct K two-class classifiers; each classifier $\text{sign}(\mathbf{w}^{(k)\top} \mathbf{x})$ discriminates class k from the others
 - Prediction: the most probable class with the largest $\mathbf{w}^{(k)\top} \mathbf{x}$
- Approach 2: One-versus-one
 - Construct $K(K - 1)/2$ two-class classifiers, each of which discriminates between a pair of two classes
 - Prediction by voting



confidence

Error Correcting Output Code (ECOC) :

An approach inspired by error correcting coding

- Approach 3: Error correcting output code (ECOC)
 - Construct a set of two-class classifiers, each of which discriminates between two groups of classes, e.g. AB vs. CD
 - Prediction by finding the nearest code in terms of Hamming distance

codes

class	two-class classification problems					
	1	2	3	4	5	6
A	1	1	1	1	1	1
B	1	-1	1	-1	-1	-1
C	-1	-1	-1	1	-1	1
D	-1	1	1	-1	-1	1
prediction	1	1	1	1	1	-1

code for class A

Design of ECOC :

Code design is the key for good classification

- Codes (row) should be apart from each other in terms of Hamming distance
 - E.g., Hadamard codes

codes

class	two-class classification problems					
	1	2	3	4	5	6
A	1	1	1	1	1	1
B	1	-1	1	-1	-1	-1
C	-1	-1	-1	1	-1	1
D	-1	1	1	-1	-1	1

Hamming distances between codes

class	A	B	C	D
A	0	4	4	3
B		0	4	3
C			0	3
D				0

Multi-class logistic regression model:

One model parameter vector for each class

- More direct modeling of multi-class classification
 - One parameter vector $\mathbf{w}^{(k)}$ for each class k : $\mathbf{w}^{(k)\top} \mathbf{x}$
 - Multi-class linear classifier: $f(\mathbf{x}) = \operatorname{argmax}_{k \in \mathcal{Y}} \mathbf{w}^{(k)\top} \mathbf{x}$
 - Multi-class logistic regression: $P(k|\mathbf{x}) = \frac{\exp(\mathbf{w}^{(k)\top} \mathbf{x})}{\sum_{k' \in \mathcal{Y}} \exp(\mathbf{w}^{(k')\top} \mathbf{x})}$
 - converts real values into positive values, and then normalizes them to obtain a probability value $\in [0,1]$

Training multi-class logistic regression: (Regularized) maximum likelihood estimation

- Find the parameters that minimizes the negative log-likelihood

$$J(\{\mathbf{w}^{(y)}\}_y) = - \sum_{i=1, \dots, N} \log p(y^{(i)} | \mathbf{x}^{(i)}) + \gamma \sum_{y \in \mathcal{Y}} \|\mathbf{w}^{(y)}\|_2^2$$

- $\|\mathbf{w}^{(y)}\|_2^2$: a regularizer to avoid overfitting

- For multi-class logistic regression $P(k|\mathbf{x}) = \frac{\exp(\mathbf{w}^{(k)\top} \mathbf{x})}{\sum_{k' \in \mathcal{Y}} \exp(\mathbf{w}^{(k')\top} \mathbf{x})}$

$$J = - \sum_i \mathbf{w}^{(k)} \top \mathbf{x}^{(i)} + \sum_i \log \sum_{k' \in \mathcal{Y}} \exp(\mathbf{w}^{(k')\top} \mathbf{x}^{(i)}) + \text{reg.}$$

–Minimization using gradient-based optimization methods

Summary:

Classification

- Classification is a supervised learning task for predicting discrete labels from input data
- Linear classifiers use a weighted sum of input features to separate classes with a hyperplane
- Loss minimization (with regularization) and probabilistic modeling (like logistic regression) are two main frameworks
- Gradient-based optimization (e.g., gradient descent, Newton's method) is used to learn model parameters
- Multi-class classification can be built from binary classifiers or modeled directly with a separate weight vector per class